⚔️ Trade-Offs & the TCD

Week 4 • Day 5 • Technical Architecture & Constraints

TPM Academy — TCD ships today

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- Run an **integration pass** across §§1–6
- Cross-review with another triad and ship the TCD as a sibling to the PRD

🕑 Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin TCD §§1–4 on the wall
09:15 – 10:30	Block 1 + Activity 1	Surface the trade-offs (8–10 candidates)
10:45 – 12:00	Block 2 + Activity 2	Write the top 5 trade-offs
13:00 – 14:15	Block 3 + Activity 3	Stakeholder sign-off matrix
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Integration + cross-review + sign-off

Block 1 — Surface the Trade-Offs

Mature architecture documents name the tensions explicitly

📚 Mature vs Immature Trade-Offs

❌ Immature

"We considered microservices but chose monolith."

Description, not trade-off.

✅ Mature

"Option A: modular monolith. Option B: extract reconcile into a service. We chose A. Accepted cost: future extraction needs coordinated refactor. Revisit if traffic exceeds 5× rest-of-app or a second team contributes."

The pattern: Option A vs Option B → choice → accepted cost → revisit trigger.

📚 Five Categories of Architectural Trade-Off

Category	Example tension
Coupling vs independence	Module in monolith vs new service
Consistency vs availability	Strong-consistency reads vs replica reads
Latency vs durability	Sync write vs async with retry
Speed vs robustness	Ship happy path now, harden later
Generality vs simplicity	Build the generic capability or just this case

Aim: top-5 trade-offs span **at least 3 categories**. All in one category usually missed real tensions.

👥 Activity 1 — Surface Trade-Offs

Triads • 35 minutes

Brainstorm 8–10 candidates from §§1–4. Categorize. Cull to 5 finalists. Rough out structure.

- 15 min: brainstorm 8–10
- 5 min: categorize
- 10 min: cull to 5
- 5 min: rough Option A / B / Choice / Cost / Trigger

⏱️ 15 + 5 + 10 + 5

Block 2 — Write the Top 5

Final TCD §5 prose

📝 The Trade-Off Template

```
### Trade-off N — <short name="">
**Tension:** <category> — <framing in="" one="" sentence="">
**Option A:** <1–2 sentences>
**Option B:** <1–2 sentences>
**Choice:** Option A.
**Why:** <2–3 sentences — defended in business + operational terms>
**Accepted cost:** <what we="" give="" up,="" concretely="">
**Revisit trigger:** <metric or="" organizational="" change=""></metric></what></framing></category></short>
```

"Accepted cost: complexity" is vague. Force concreteness: *maintenance burden, debugging difficulty, on-call exposure*.

📝 Worked Trade-Off

```
### Trade-off 2 – Sync vs async write to audit
**Tension:** Latency vs durability.
**Option A:** Synchronous write to audit event store
before responding to user.
**Option B:** Async publish to Kafka audit topic;
consumer writes to store.
**Choice:** Option B.
**Why:** Synchronous would add 30–80ms to the reconcile
latency budget, breaking the p95 ≤ 1000ms SLO. Async
with Kafka durability is sufficient for SOC 2 audit
requirements. The DLQ on the consumer handles rare
write failures.
**Accepted cost:** Audit events may lag 1–10s behind user
action; a regulator querying within seconds may see
stale state.
**Revisit trigger:** A regulator or customer requires
synchronous audit visibility within < 1 second.
```

👥 Activity 2 — Write the Top 5

Triads • 40 minutes

- 15 min: solo drafts (each member writes 2)
- 15 min: pool + refine
- 10 min: consistency check — do trade-offs contradict each other?

⏱️ 15 + 15 + 10

Block 3 — Stakeholder Sign-Off Matrix

A constraint without an owner is a constraint nobody implements

📋 The Sign-Off Matrix

#	Constraint (link)	Stakeholder	Status	Next step
1	§1 Architecture stance	Lin (Architect)	Proposed	30-min walkthrough
2	§3 Threat model	Pat (Security lead)	Proposed	Send brief + 30-min walk
3	§3 SOC 2 retention	Sam (Compliance)	Proposed	Email + 15-min sync
4	§4 Latency SLO	Kim (Eng lead)	Discussed	Confirm sprint review
5	§4 Rate-limit policy	Jamie (Platform)	Proposed	Slack + review
6	§5 Trade-off 1	Lin (Architect)	Proposed	Same as #1
7	SSO scope addition	Casey (Identity)	Confirmed	None

🎯 Status Values (be honest)

Status	Meaning
Proposed	TPM has drafted; stakeholder hasn't reacted
Discussed	Conversation happened; no formal sign-off
Approved	Stakeholder said yes (capture how)
Blocked	Pushback; needs different proposal

> **Most entries should be Proposed today.** That's honest. The matrix is the action list for Week 6.

👥 Activity 3 — Sign-Off Matrix

Triads • 40 minutes

- 15 min: list every constraint needing sign-off
- 10 min: assign a real person per row
- 10 min: set the "next step" for each
- 5 min: status sanity check (most should be Proposed)

⏱️ 15 + 10 + 10 + 5

Block 4 — Integration + Cross-Review + Sign-Off

Read top to bottom; pair-review; ship

✅ The Integration Checklist

Check	Pass criterion
§1 stance ↔ §2 diagrams	Stance shows as expected container layout
§2 integrations ↔ §3 threats	Every boundary covered or explicitly out-of-scope
§3 NFRs ↔ §4 SLOs	Conflicts surface as §5 trade-offs
§4 SLOs ↔ §5 trade-offs	Latency cost appears in §5 if accepted
§§1–5 ↔ §6 stakeholders	Every load-bearing decision has an owner
No fortune-cookie prose	Specific to this feature

🔎 Cross-Review Focus

Reviewer rubric (for scoring):

Dimension	Weight
Stance defensibility	20%
Threat-model rigor	20%
Component clarity	15%
SLO realism	20%
Trade-off honesty	15%
Stakeholder map	10%

🔎 Cross-Review Focus

1. **Is each trade-off really a trade-off**, or just a description?

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1. **Is each trade-off really a trade-off**, or just a description?
2. **Does the stakeholder matrix have obvious gaps?**

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🔎 Cross-Review Focus

1. **Is each trade-off really a trade-off**, or just a description?
2. **Does the stakeholder matrix have obvious gaps?**
3. **Are SLOs and trade-offs internally consistent?**

> **Reviewer rubric (for scoring):**

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Component clarity	15%
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👥 Activity 4 — Integration + Cross-Review + Sign-Off

Triads • 45 minutes • Wrap

- 10 min: integration pass (solo + pool)
- 20 min: cross-review with another triad (rubric)
- 15 min: adopt / defer / reject + revise + internal sign-off

⏱️ 10 + 20 + 15 · 15-min wrap-share

🎉 Week 4 Wrap

What we shipped this week

- Architecture stance with revisit trigger
- Integration map + C4 diagrams
- STRIDE threat model + revised security NFRs
- Three SLOs + latency budget + rate limits
- Top 5 trade-offs
- Stakeholder sign-off matrix
- Integrated TCD — sibling to PRD

What opens Week 5

- **Technical Infrastructure & Modeling**
- Databases & data logic
- Cloud architecture
- REST & SOAP APIs
- Performance baselines & monitoring
- TCD becomes the spine

Bring to Monday: Your TCD + your PRD. The TCD's component map drives Week 5's data modeling; the SLOs drive performance baselines.

💬 Questions & Reflection

Which of your trade-offs would you change if a senior architect spent 30 minutes with you?